Aniketh Parkala

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Research Interests

Nonlinear and adaptive control · unconventional and fault-tolerant aerial vehicles · control under actuator delay · experimental robotics · mechanism-controller co-design

Education

IIIT Hyderabad — Dual Degree: B.Tech ECE + MS by Research in Robotics Nov 2021 – Dec 2026 (expected)
Advisor: Prof. Harikumar Kandath *MS coursework GPA 9.0/10*
Coursework: Robot Dynamics & Control, Advances in Robotics and Control, Mobile Robotics, Mechatronics System Design, Intro. to UAV Design, Reinforcement Learning, Statistical Methods in AI

Publications

- **A. Parkala** and H. Kandath, "Spinning Quadrotor: Hover Thrust Augmentation with Passive Lifting Surfaces," *Proc. Int. Conf. on Unmanned Aircraft Systems (ICUAS)*, 2026. **Best Paper Award Finalist.** DOI | Project page | Code

Research Experience

Research Assistant, Robotics Research Center, IIIT Hyderabad May 2023 – Present
On-board control of a spinning quadrotor at high spin rates *T-Mech, in preparation*

- Identified **actuator delay as the limiting factor in high-rate spinning flight** via frequency-domain analysis of spin rate and control-axis phase error.
- Developed a **delay-compensated spin controller** combining an **adaptive phase compensator** with **rotor-dynamics inversion**; together they enable stable simulated flight at spin rates where either alone fails
- Integrated the controller into **PX4** (custom control allocation, spin-invariant pilot modes, rotor-failure injection) and corrected **EKF2 divergence under rapid rotation**.
- Designed and prototyped the airframe and wings solo; sized the wings for lift in the spinning regime with BEMT, CFD and XFLR5
- Built an **Isaac Sim/Pegasus + PX4 SITL** sim-to-hardware pipeline; identified dynamics from flight logs and achieved **fully controlled spinning flight at 15 rad/s** on a Kakute H7 platform with motion capture.

Hover thrust augmentation with passive lifting surfaces *ICUAS 2026, Best Paper Finalist*

- Designed passive wings and showed on hardware that they cut hover thrust by 22% during steady spin at around 8–10 rad/s.

Teaching Experience

TA, Principles of Perception and Planning for Aerial Vehicles (NPTEL) July 2025 – September 2026
Developed all lecture slides, assignments and quizzes for the course, working to the guidelines of three faculty members. Delivered 3 hours of hands-on tutorial lectures.

TA, Introduction to UAV Design Jan – May 2025
Taught PX4, ROS 2, and XFLR5 to undergraduate, Master's, and PhD students.

TA, Digital Systems and Microcontrollers Aug – Dec 2023, 2024
Ran Arduino and digital-circuits labs for 340+ students across two offerings.

Selected Projects

Adaptive Sliding-Mode Control for Bipedal Locomotion *(ROS 2, Gazebo)*

- Developed joint-space PID and effort-based Adaptive Sliding Mode Control for gait control under model uncertainty; generated and evaluated a full walking gait cycle in simulation.

CanSat 2024, International Competition 2023 – 2024

- Designed mechanical subsystems and structure; 4th worldwide at Preliminary Design Review, 30th in world finals.

Awards

KVPY Fellow, DST, Govt. of India (2019–20) · CCMB Young Innovators Programme, Hyderabad (2019) · CanSat 2024 PDR: 4th worldwide · ICUAS 2026 Best Paper Finalist

Technical Skills

Control	Nonlinear, adaptive, sliding-mode; LQR; backstepping; Lyapunov analysis; dynamic inversion; control allocation; system identification
Flight stacks	PX4 (controllers, EKF2, allocation, uORB); PX4 SITL; ROS 2; MAVLink; QGroundControl; ULog
Simulation	Isaac Sim / Pegasus; MuJoCo; Gazebo; ANSYS Fluent; XFLR5
Programming	C/C++; Python; MATLAB; Bash; Git; Linux
Hardware	Kakute H7; ESP32; motors/ESCs; motion capture; PCB design; Verilog; SPICE; Fusion 360